From chat to measured results.

Discover. Evaluate. Choose with evidence.

ModelPicker turns a Claude conversation into a reproducible CPU benchmark of text classifiers. Discover Hugging Face models, import labeled data, compare quality and speed, and export results. Supports single-label sentiment, spam and ticket classification.

23	5	1,000	CPU
MCP tools	models per plan	rows per model	local inference

Discover & import

Search models and datasets; inspect labels and revisions. Import CSV, JSONL, JSON, pasted rows, trusted URLs or Hub datasets.

Plan & evaluate

Seeded sampling, pinned revisions and identical rows. Background jobs with progress, cancellation and resource limits.

Measure & reuse

Accuracy, macro F1, per-label scores, latency, throughput and memory. Save runs; export reports and Python integration code.

Python · MCP / FastMCP · PyTorch · Transformers · Hugging Face · SQLite

01 / LOCAL CONNECTION — INSTALL ON YOUR OWN COMPUTER

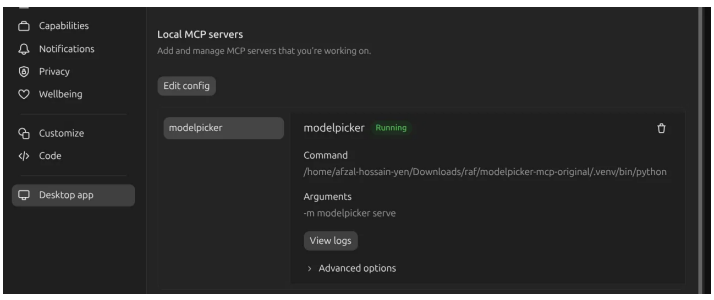
Requires Git, uv, Python and Claude Desktop. The helper installs the Python 3.11 CPU runtime.
1. `git clone https://github.com/samiremon/modelpicker-mcp.git`
2. `cd modelpicker-mcp`, then `python3 scripts/setup_local.py` (Windows: `python`).
3. Open generated `~/.modelpicker/desktop-config.json`; merge its `mcpServers.modelpicker` entry into Claude → Desktop app → Local MCP servers → Edit config.
4. Fully quit and reopen Claude; enable ModelPicker. Uses local CPU / stdio; no OAuth.
Full installation guide, Windows paths and dataset-directory options

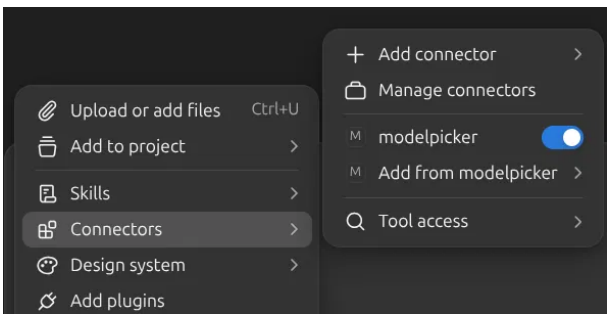
02 / REMOTE CONNECTION

1. Install ModelPicker on the host as above, then start a public HTTPS tunnel:
`cloudflared tunnel --url http://127.0.0.1:7860`
2. Use the tunnel's HTTPS origin to start the server (one command):
`uv run --no-sync modelpicker serve --transport streamable-http --public-url https://YOUR-TUNNEL-HOST`
3. Claude Settings → Connectors → Add custom connector:
`https://YOUR-TUNNEL-HOST/mcp`. Authorize in the browser using `~/.modelpicker/owner-password.txt`. Keep the server and tunnel running.

Uses the server host's CPU. Quick-tunnel URLs change on restart; reconnect after server restarts. One trusted owner per installation. Use one HTTP service for multiple clients; avoid concurrent servers sharing storage.

CONNECTED IN CLAUDE DESKTOP





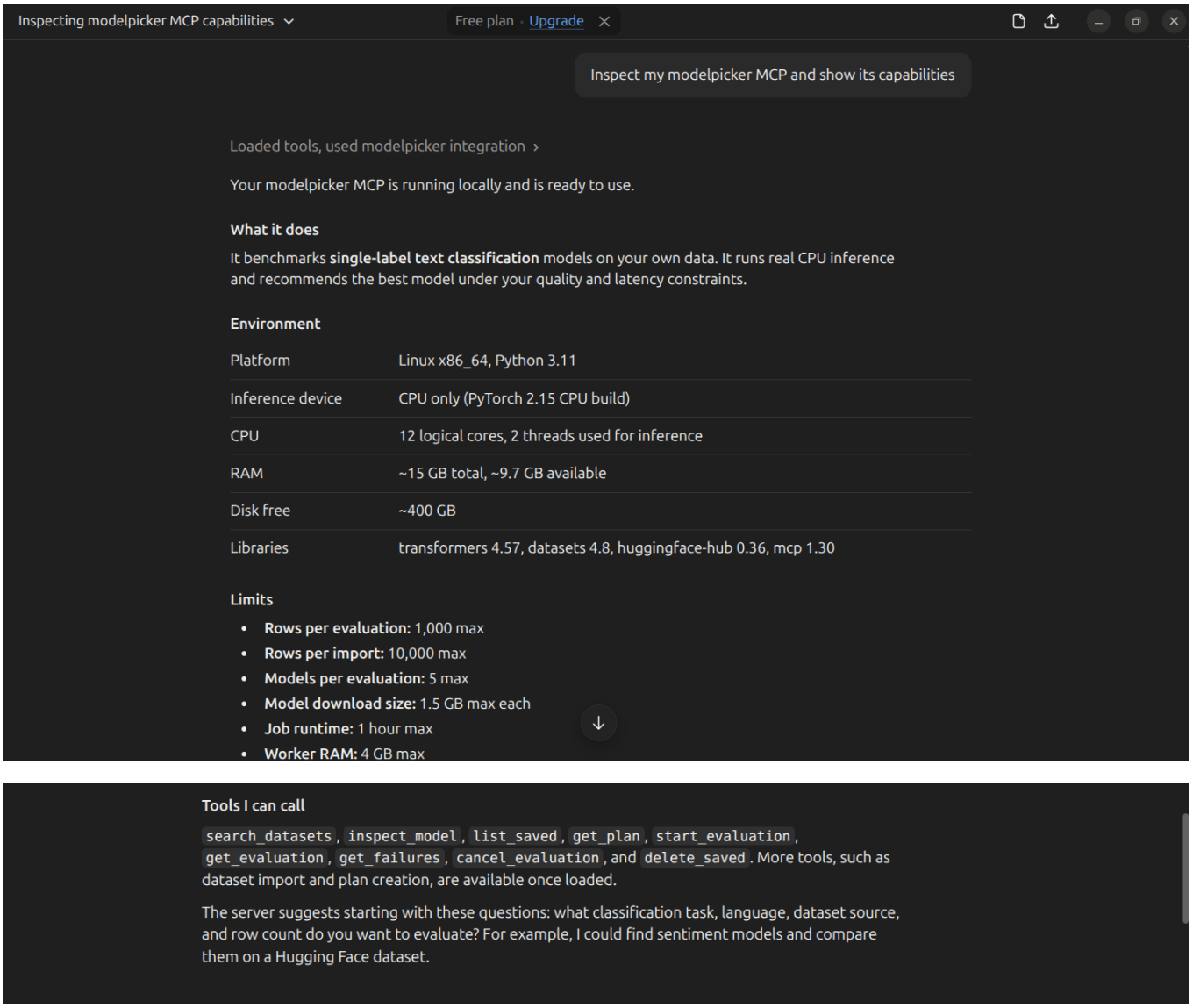
Text and tool results sent to Claude reach the provider. No paid inference API is required; downloads and hardware costs still apply.

01 / Inspect capabilities

“Inspect my modelpicker MCP and show its capabilities”

Implemented workflow tools: get_capabilities

View conversation



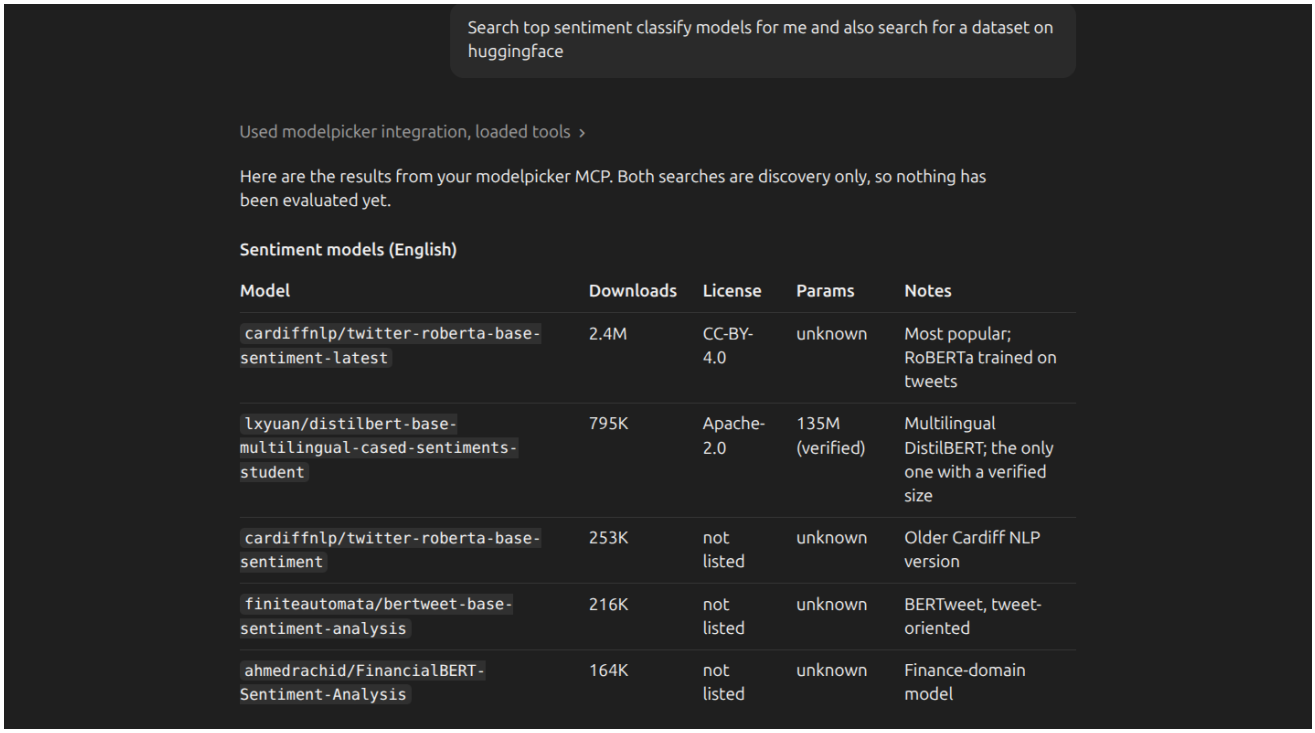
The session reports the local CPU environment, package versions and resource limits. These are historical observations from this session. Every tool name visible in the screenshot exists in the project.

02 / Discover models & datasets

“Search top sentiment classify models for me and also search for a dataset on huggingface”

Implemented workflow tools: search_models · search_datasets

View conversation



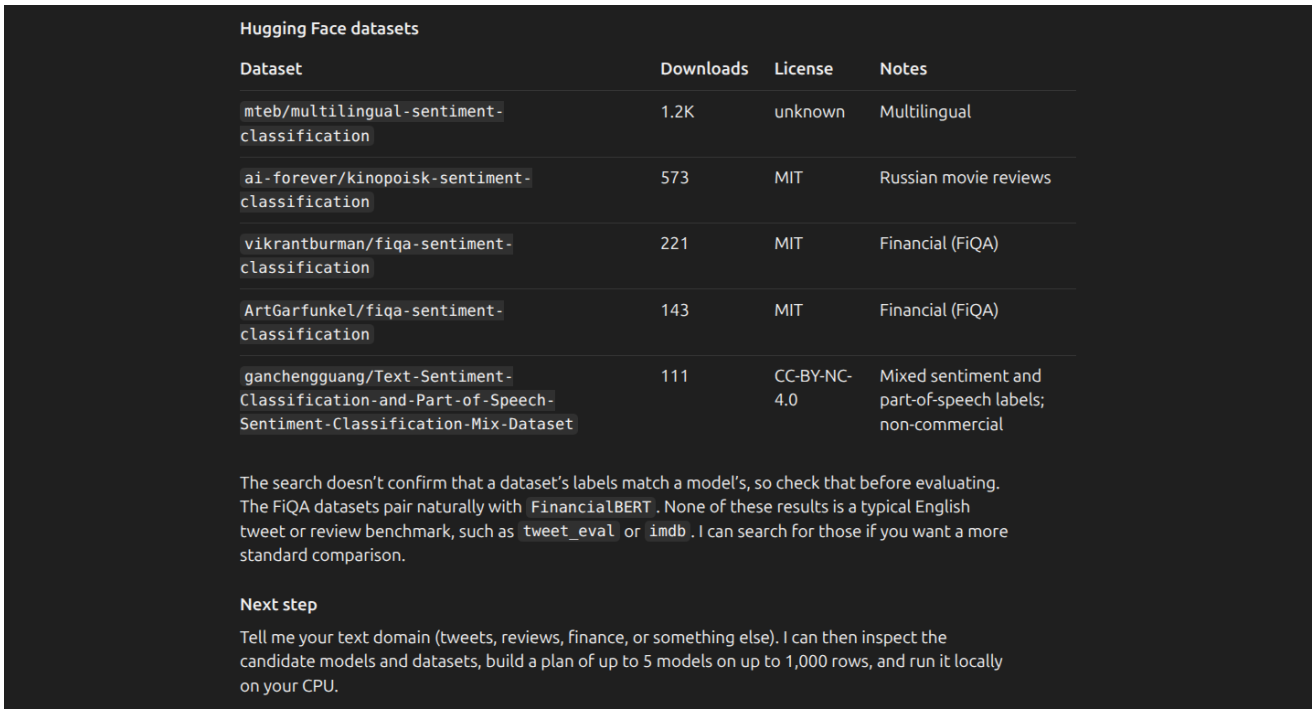
Search top sentiment classify models for me and also search for a dataset on huggingface

Used modelpicker integration, loaded tools ›

Here are the results from your modelpicker MCP. Both searches are discovery only, so nothing has been evaluated yet.

Sentiment models (English)

Model	Downloads	License	Params	Notes
`cardiffnlp/twitter-roberta-base-sentiment-latest`	2.4M	CC-BY-4.0	unknown	Most popular; RoBERTa trained on tweets
`lxyuan/distilbert-base-multilingual-cased-sentiments-student`	795K	Apache-2.0	135M (verified)	Multilingual DistilBERT; the only one with a verified size
`cardiffnlp/twitter-roberta-base-sentiment`	253K	not listed	unknown	Older Cardiff NLP version
`finiteautomata/bertweet-base-sentiment-analysis`	216K	not listed	unknown	BERTweet, tweet-oriented
`ahmedrachid/FinancialBERT-Sentiment-Analysis`	164K	not listed	unknown	Finance-domain model



Hugging Face datasets

Dataset	Downloads	License	Notes
`mteb/multilingual-sentiment-classification`	1.2K	unknown	Multilingual
`ai-forever/kinopoisk-sentiment-classification`	573	MIT	Russian movie reviews
`vikrantburman/fiqa-sentiment-classification`	221	MIT	Financial (FiQA)
`ArtGarfunkel/fiqa-sentiment-classification`	143	MIT	Financial (FiQA)
`ganchengguang/Text-Sentiment-Classification-and-Part-of-Speech-Sentiment-Classification-Mix-Dataset`	111	CC-BY-NC-4.0	Mixed sentiment and part-of-speech labels; non-commercial

The search doesn’t confirm that a dataset’s labels match a model’s, so check that before evaluating. The FiQA datasets pair naturally with `FinancialBERT`. None of these results is a typical English tweet or review benchmark, such as `tweet_eval` or `imdb`. I can search for those if you want a more standard comparison.

Next step

Tell me your text domain (tweets, reviews, finance, or something else). I can then inspect the candidate models and datasets, build a plan of up to 5 models on up to 1,000 rows, and run it locally on your CPU.

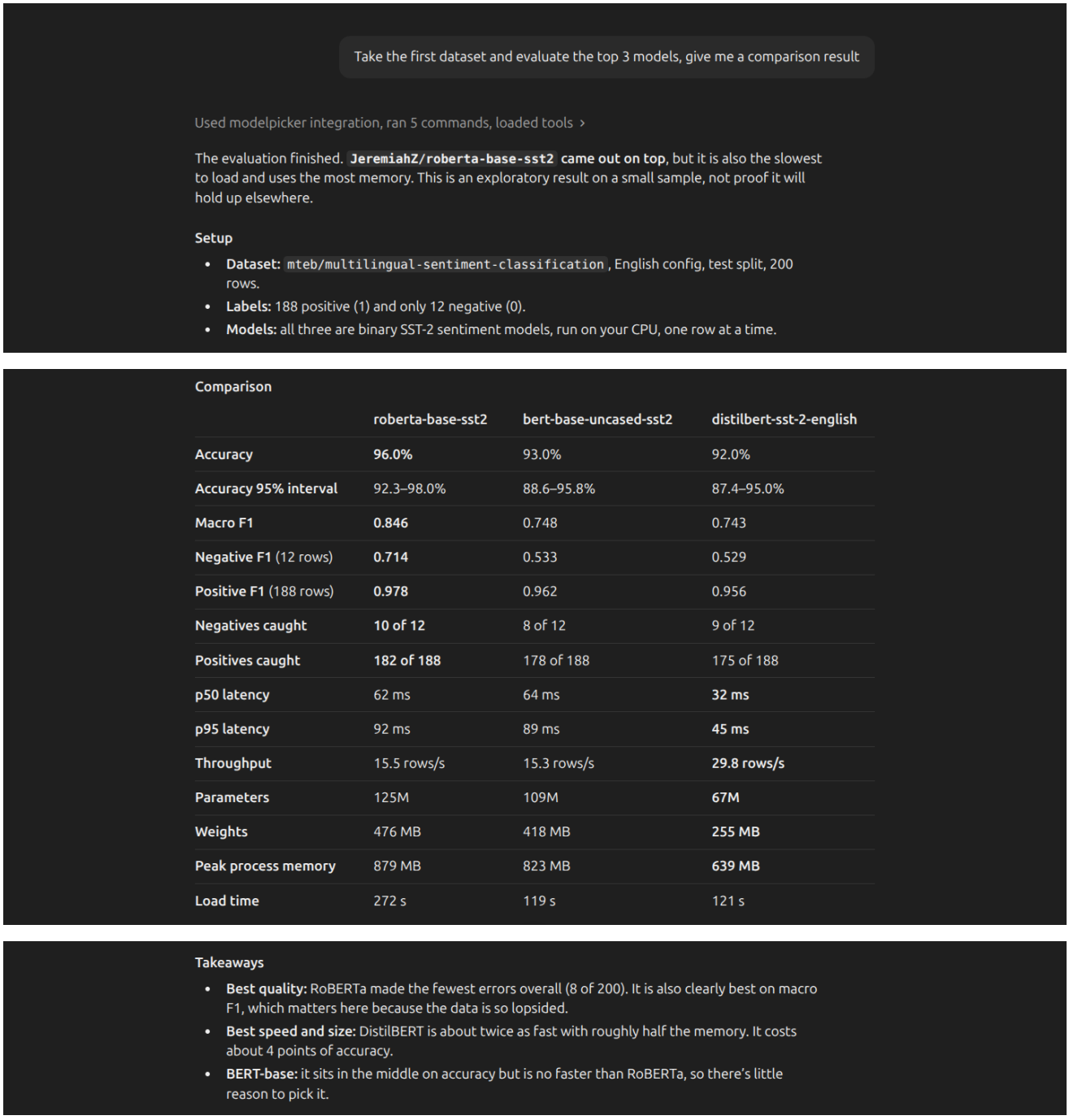
These are discovery results, not an evaluated ranking. Download counts, licenses and model metadata reflect the recorded session; compatibility requires inspection before evaluation.

03 / Evaluate & compare

“Take the first dataset and evaluate the top 3 models, give me a comparison result”

Implemented workflow tools: inspect_huggingface_dataset · import_huggingface_dataset · inspect_model · create_evaluation_plan · start_evaluation · get_evaluation

View conversation



Verified against the saved evaluation: 200 English test rows (188 positive, 12 negative), 600 predictions, no execution errors. Accuracy: RoBERTa 96%, BERT 93%, DistilBERT 92%. Imbalanced, exploratory sample. Inference latency/throughput exclude download, loading and warmup; memory and weight figures use MiB.